

TECHNICAL ASSESSMENT

Image Restoration System — Design Brief

For the role of Computer Vision Engineer

ISSUED BY	Tata Elxsi — AI & Vision Systems Group
ROLE	Computer Vision Engineer (Product Engineering)
ASSESSMENT TYPE	Take-home design and implementation exercise
DATE OF ISSUE	06 May 2026
TIME ALLOWED	48 hours from receipt

1. CONTEXT

The candidate is being evaluated for a Computer Vision role at Tata Elxsi, where engineering teams build production-grade visual systems for a range of product and platform clients. This assessment presents a problem representative of the work the team undertakes, and is intended to evaluate the candidate's ability to design, implement, and clearly communicate a working solution.

2. PROBLEM STATEMENT

Design and implement a system that restores blurred, noisy, or otherwise low-quality images to produce a clean, visually improved output of the same scene.

3. OBJECTIVE

Given an input image that exhibits one or more of the following characteristics:

- (a) blur arising from motion, defocus, or other sources;
- (b) noise of unknown distribution; or
- (c) general low quality due to compression or sensor limitations,

the system shall produce a corresponding output image that is perceptibly cleaner, sharper, and faithful to the underlying scene.

4. FUNCTIONAL REQUIREMENTS

4.1 Image Type Support

The system shall correctly accept and process both grayscale (single-channel) and colour (three-channel RGB) images.

4.2 Processing Pipeline

The system shall accept an arbitrary input image, perform the necessary processing internally, and return a single enhanced output image without requiring additional user intervention.

4.3 Quality of Output

The output image shall:

- (a) exhibit a measurable reduction in blur relative to the input;
- (b) demonstrate improved overall visual clarity; and
- (c) preserve important structural and semantic features of the original scene, without introducing visible artefacts or hallucinated content.

4.4 Generalisation

The system shall not be tuned to, or restricted to, any single dataset. It is expected to perform reasonably on previously unseen images drawn from a variety of distributions.

5. INPUT AND OUTPUT SPECIFICATION

Input	A single blurred, noisy, or low-quality image, in any common raster format (e.g. JPEG, PNG).
Output	The corresponding restored image, of the same dimensions, saved in a standard raster format.

6. EVALUATION CRITERIA

Submissions will be assessed against the following criteria. Candidates are advised to address each criterion explicitly in their submission.

Criterion	Description
6.1 Output Quality	The visual quality of the restored image, evaluated both qualitatively and where appropriate using standard quantitative metrics.
6.2 Robustness	The ability of the system to handle different types and severities of blur and noise.
6.3 Generalisation	Performance on inputs not seen during development or training.
6.4 Clarity of Approach	The quality, structure, and correctness of the candidate's explanation of their methodology.

7. SUBMISSION REQUIREMENTS

Candidates are required to submit, in addition to the working system, a short explanation video. The video is a mandatory component of this assessment and shall cover, at a minimum:

- (a) the candidate's overall approach to the problem and the rationale for the chosen technique;
- (b) a clear description of how the system works internally, including any pre-processing, model or algorithmic stages, and post-processing; and
- (c) a step-by-step dry run on at least one representative input image, demonstrating how the input is transformed at each stage of the pipeline.

Submissions that do not include the explanation video will be considered incomplete.

8. TIMELINE

The assessment shall be completed and submitted within **forty-eight (48) hours** from the time of issue. Late submissions will not be considered unless an extension has been agreed in writing in advance.

Note to candidate: You are encouraged to use any publicly available libraries, frameworks, or pre-trained models, provided that all such usage is disclosed in your explanation. Originality of approach is valued, but so is engineering judgement and clarity of communication. If any requirement in this document is unclear, please request clarification before commencing work.